

The background features a large white circle in the center, partially overlapping a light blue rectangle on the left and a light pink rectangle on the right. A large, dark blue shape, resembling a stylized 'U' or a wide smile, is positioned at the bottom, framing the white circle.

LIBRARY MANAGEMENT SYSTEM PROJECT

OBJECTIVE

The objective of this project is to design and implement a relational database for a library management system using MySQL. The database will manage information on about books, library members, and borrowing activities. It will also allow performing various queries to retrieve insights and support library operations.

1. RETRIEVE ALL BOOK DETAILS

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- `select * from Books;`

Result Grid						
		Filter Rows:	Edit:		Export/Import:	Wrap Cell Content:
	Book_id	Title	Author	Genre	PublishedYear	CopiesAvailable
▶	1	To Kill a Mockingbird	Harper Lee	Fic on	1960	5
	2	1984	George Orwell	Dystopian	1949	3
	3	Moby Dick	Herman Melville	Adventure	1851	4
	4	The Great Gatsby	F. Sco Fitzgerald	Classic	1925	2
	5	Pride and Prejudice	Jane Austen	Romance	1813	6
	6	The Catcher in the Rye	J.D. Salinger	Fic on	1951	4
	7	The Hobbit	J.R.R. Tolkien	Fantasy	1937	7
	8	The Da Vinci Code	Dan Brown	Thriller	2003	5
	9	The Alchemist	Paulo Coelho	Philosophy	1988	6
	10	War and Peace	Leo Tolstoy	Historical Fic on	1869	3
•	NULL	NULL	NULL	NULL	NULL	NULL

2. FIND ALL MEMBERS WHO JOINED AFTER JANUARY 1, 2022

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- `select * from Members where MembershipDate > '2022-01-01';`

Result Grid   Filter Rows: Edit:    Export/Import:   Wrap Cell Content: 					
	Member_id	Name	MembershipDate	Email	PhoneNumber
▶	1	Alice Johnson	2022-01-15	alice@example.com	123-456-7890
	2	Bob Smith	2023-03-10	bob@example.com	234-567-8901
	5	Evan Parker	2022-06-30	evan@example.com	567-890-1234
	6	Frank Castle	2023-01-01	frank@example.com	678-890-2345
	7	Grace Hopper	2022-11-30	grace@example.com	789-012-3456
	9	Isaac Newton	2023-04-05	isaac@example.com	901-234-5678
	10	Jack Ryan	2022-03-22	jack@example.com	012-345-6789
★	NULL	NULL	NULL	NULL	NULL

3. LIST THE BOOKS THAT HAVE FEWER THAN 3 COPIES AVAILABLE

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- `select * from books where CopiesAvailable < 3;`

Result Grid						
		Filter Rows:		Edit:		Export/Import:
						Wrap Cell Content:
	Book_id	Title	Author	Genre	PublishedYear	CopiesAvailable
▶	4	The Great Gatsby	F. Sco Fitzgerald	Classic	1925	2
•	NULL	NULL	NULL	NULL	NULL	NULL

4. SHOW THE BORROWING HISTORY ALONG WITH MEMBER NAMES AND BOOK TITLES

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- select
B.Borrow_id,M.Name,BK.Title,B.BorrowDate,B.ReturnDateFrom
BorrowingTable B Join Members M ON B.Member_id =
M.Member_id Join Books BK ON B.Book_id = BK.Book_id;

Result Grid Filter Rows: Export: Wrap Cell Content:					
	Borrow_id	Name	Title	BorrowDate	ReturnDate
▶	1	Alice Johnson	To Kill a Mockingbird	2023-01-10	2023-01-20
	2	Bob Smith	1984	2023-02-15	2023-02-25
	3	Charlie Brown	Moby Dick	2023-03-05	NULL
	4	Diana Prince	The Great Gatsby	2023-04-01	2023-04-15
	5	Evan Parker	Pride and Prejudice	2023-05-20	NULL
	6	Frank Castle	The Catcher in the Rye	2023-01-10	2023-01-20
	7	Grace Hopper	The Hobbit	2023-02-05	2023-02-15
	8	Hannah Baker	The Da Vinci Code	2023-03-10	NULL
	9	Isaac Newton	The Alchemist	2023-04-20	2023-05-01
	10	Jack Ryan	War and Peace	2023-05-25	NULL

5. IDENTIFY OVERDUE BOOKS (NOT RETURNED BY TODAY'S DATE)

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- select BK.Title,M.Name,B.BorrowDateFrom BorrowingTable B
Join Books BK on B.Book_id = BK.Book_id Join Members M on
B.Member_id = M.Member_id where B.ReturnDate is NULL
AND B.BorrowDate < curdate();

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Title	Name	BorrowDate
▶	Moby Dick	Charlie Brown	2023-03-05
	Pride and Prejudice	Evan Parker	2023-05-20
	The Da Vinci Code	Hannah Baker	2023-03-10
	War and Peace	Jack Ryan	2023-05-25

2023-05-25

6. SHOW THE TOP 3 MOST BORROWED BOOKS

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- SELECT BK.Book_id,BK.Title,COUNT(*) AS TotalBorrowed
FROM BorrowingTable B JOIN Books BK ON B.Book_id =
BK.Book_id GROUP BY BK.Book_id, BK.Title ORDER BY
TotalBorrowed DESC LIMIT 3;

Result Grid	Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
Book_id	Title	TotalBorrowed		
1	To Kill a Mockingbird	1	1	
2	1984	1		
3	Moby Dick	1		

7. DISPLAY THE TOTAL NUMBER OF BOOKS IN EACH GENRE

9

- `SELECT Genre,COUNT(*) AS TotalBooks FROM Books GROUP BY Genre;`

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	Genre	TotalBooks			
▶	Fic on	2			
	Dystopian	1			
	Adventure	1			
	Classic	1			
	Romance	1			
	Fantasy	1			
	Thriller	1			
	Philosophy	1			
	Historical Fic on	1			

8. RETRIEVE THE BORROWING HISTORY FOR A SPECIFIC MEMBER (MEMBER_ID = 3)

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- SELECT M.Name,BK.Title,B.BorrowDate,B.ReturnDate FROM BorrowingTable B JOIN Members M ON B.Member_id = M.Member_id JOIN Books BK ON B.Book_id = BK.Book_id WHERE M.Member_id = 3;

Result Grid				
Filter Rows:				
Export:				
Wrap Cell Content: ☐				
	Name	Title	BorrowDate	ReturnDate
▶	Charlie Brown	Moby Dick	2023-03-05	NULL

9. CALCULATE THE AVERAGE NUMBER OF DAYS BOOKS ARE BORROWED BEFORE BEING RETURNED

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- `SELECT AVG(DATEDIFF(ReturnDate, BorrowDate)) AS AverageBorrowDays FROM BorrowingTable WHERE ReturnDate IS NOT NULL;`

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	AverageBorrowDays			
▶	10.8333			

10. FIND THE LEAST BORROWED BOOK

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- SELECT BK.Book_id,BK.Title,COUNT(B.Borrow_id) AS TimesBorrowed FROM Books BK LEFT JOIN BorrowingTable B ON BK.Book_id = B.Book_id GROUP BY BK.Book_id, BK.Title ORDER BY TimesBorrowed ASC LIMIT 1;

Result Grid			
Filter Rows:			
Export: 			
Wrap Cell Content: 			
Fetch rows: 			
	Book_id	Title	TimesBorrowed
▶	1	To Kill a Mockingbird	1

**THANK
YOU**